

Local Language-Model Inference with EDSL: An Exploratory Behavioral Evaluation of Bonsai 2 27B

Technical research note

September 19, 2026

Abstract

This note documents local execution of PrismML’s Ternary Bonsai 2 27B through EDSL and an exploratory evaluation of its responses to behavioral-decision problems. A compressed GGUF model was served by PrismML’s llama.cpp runtime on an Apple M4 Max with 48 GB of unified memory. The evaluation comprised 84 one-question interviews spanning 11 test families and 21 conditions, with four responses per condition. The model matched the answer key in 41 of 44 objectively scored responses. All three errors occurred in a newly written conjunction vignette: the model selected the conjunction in three of four trials, while answering the canonical Linda item correctly in all four. Most paired preference tasks showed no observed bias-direction shift, although the sample is small and some tasks have strong cues or ceiling effects. The report provides inference settings, EDSL implementation details, response audits, and the complete prompt inventory. It also assesses, without deploying, how the model could be served from Modal for Macaw. The findings establish practical local integration and identify a useful generalization failure; they do not establish general rationality or human-like behavior.

1 Purpose and evidentiary scope

Two questions motivated this exercise. First, could a recently released, compressed language model run usefully on a MacBook Pro and answer EDSL questions? Second, would its choices exhibit familiar departures from normative reasoning? The first question concerns an implemented and tested system. The second is addressed by a small, deliberately selected battery, not a validated psychometric instrument or a comparison with human respondents.

The battery draws on research on heuristics and probability judgments ([Tversky and Kahneman, 1974, 1983](#)), equivalent descriptions of risky choices ([Tversky and Kahneman, 1981](#)), sunk costs ([Arkes and Blumer, 1985](#)), defaults and status-quo effects ([Samuelson and Zeckhauser, 1988](#)), asymmetric dominance ([Huber et al., 1982](#)), intertemporal choice ([Frederick et al., 2002](#)), logical falsification ([Wason, 1968](#)), and intuitive arithmetic ([Frederick, 2005](#)). These citations identify the research traditions; most prompts below are adaptations rather than replications of the original instruments.

Inference took place on September 17, 2026 in the America/New_York time zone. This report was prepared from saved execution specifications and Results packages, without generating additional model responses. The factual claims about local performance come from those artifacts ([EDSL local experiment, 2026](#)). Vendor performance claims and the proposed Modal deployment are identified separately.

2 Model and local execution environment

The tested artifact was `Ternary-Bonsai-2-27B-PQ2_0.gguf`, retrieved from `prism-ml/Ternary-Bonsai-2-27B-gguf`. Its actual file size was 7,206,168,928 bytes (7.21 decimal GB). This is the two-bit-slot packing; it should not be confused with the approximately 5.95 GB densely packed ternary variant advertised for the same model family (PrismML, 2026a). File size is not a measurement of resident RAM or peak inference memory.

The runtime was PrismML’s macOS ARM64 build of `llama.cpp`, release `prism-b10683-d8f26ee`, using Metal GPU acceleration. At the time of inspection, the project required its fork for the Bonsai 2 weight formats and activation transforms (PrismML, 2026b). Both downloaded archives and weights were checked against their published SHA-256 digests. The model revision and weight digest are recorded in Appendix A. No vision projector was loaded; this experiment evaluates text only.

Table 1: Executed local configuration.

Component	Setting
Machine	MacBook Pro; Apple M4 Max; 48 GB unified memory
EDSL	Version 1.0.8.dev1
Endpoint	http://127.0.0.1:8087/v1 ; alias <code>bonsai-2-27b</code>
Serving	One slot; 8,192-token context; GPU layers requested: 999; flash attention on
Sampling	Temperature 1.0; top- p 0.95; server top- k 20; min- p 0; repetition penalty 1.0
Response limits	2,048 maximum output tokens; configured reasoning budget 512 tokens
Reasoning format	Separate reasoning field requested with <code>deepseek</code> format
Client execution	One concurrent interview; 300-second API timeout for the battery

Initial smoke tests returned “local works,” selected 4 for $2 + 2$, and answered the free-text capital-of-France question with “Paris.” Server timing lines for the initial two-question smoke test showed approximately 31.8–33.6 generated tokens per second. These are short-request observations, not a controlled throughput benchmark. The 84 retained battery completions reported 15,816 prompt tokens and 40,091 completion tokens in aggregate. Completion counts include reasoning where the server counts it; they are not counts of user-visible answer text alone. Reported creation timestamps spanned roughly 25.8 minutes, including the checkpoint/resume interval. This interval is not a direct measurement of pure model compute time. Peak RAM, energy use, and thermal behavior were not instrumented.

3 EDSL integration and execution controls

3.1 Object composition and transport

EDSL constructs a model using its `openai_compatible` inference service, pointing to the local server’s OpenAI-compatible chat-completions endpoint. A `QuestionMultipleChoice` or `QuestionNumerical` provides response validation. `ScenarioList` supplies the independent prompt conditions, and `Jobs` combines the question, scenarios, and model. Completed `Results` packages preserve parsed answers, rendered prompts, validation indicators, and raw model responses.

The composition pattern used by the battery is illustrated below. The launcher context manager starts the server if needed, waits for health readiness, and terminates only a server that it owns. An

already running server is reused and left running. The complete runner also checkpoints each test family.

Listing 1: EDSL execution pattern for a multiple-choice family.

```
from edsl import (
    Cache, Model, QuestionMultipleChoice, ScenarioList
)
from edsl.config import CONFIG
from ask_france import bonsai_server

model = Model(
    "bonsai-2-27b",
    service_name="openai_compatible",
    base_url="http://127.0.0.1:8087/v1",
    temperature=1.0, top_p=0.95, max_tokens=2048,
)
question = QuestionMultipleChoice(
    question_name="response",
    question_text="{{ prompt }}" + "\n"
        "Give a brief explanation in at most two sentences.",
    question_options="{{ options }}",
    include_comment=True,
)
# scenarios contains each condition in forward and reverse option order.
job = question.by(ScenarioList(scenarios)).by(model)
job.save("example.jobs.ep")
CONFIG.EDSL_API_TIMEOUT = "300"
with bonsai_server():
    results = job.run(
        disable_remote_inference=True,
        disable_remote_cache=True,
        fresh=True, cache=Cache(), n=2,
        max_concurrency=1, progress_bar=False,
        stop_on_exception=False,
    )
results.save("example.results.ep")
# The battery checks count, missing answers, and completion metadata.
```

Numerical families substitute `QuestionNumerical`, use one scenario per condition, and set `n=4`. Multiple-choice families use two option orders and `n=2`, yielding four answers per condition. Generation was stochastic; no model-sampling seed was fixed. The seed 20260917 controlled only the shuffle of the scenario list within each family.

3.2 Server lifecycle, failures, and caching

Three implementation lessons are material to interpreting and reproducing the experiment. First, model storage and model execution are distinct: a saved weight file does not imply a live server. An early standalone script returned a missing answer when its previously launched server was no longer running. The revised helper performs a readiness check and manages a background *subprocess*. It does not load inference into a Python thread.

Second, EDSL's default failure handling can yield a Results row with a missing answer rather than terminating the user script immediately. The one-question example was changed to raise on inference failure. The battery instead saves each family's results before rejecting incomplete output, so successful work can be inspected if another interview fails. All final 84 answers were present. The

request timeout was increased because long reasoning responses can exceed the short smoke test’s latency.

Third, `fresh=True` alone was not relied on to guarantee distinct local draws. Inspection of the local cache path showed that prompt and iteration participate in cache keys. Each family therefore used an empty in-memory `Cache()`, with EDSL iterations distinguishing repeated prompts. Both remote inference and remote caching were disabled. Saved completion IDs were checked for duplicates: there were 84 distinct IDs for 84 retained answers. Cache-hit flags were absent in some serialized data, so absence of a positive flag was not treated as sufficient evidence by itself. Server-side prompt/KV caching is compatible with generating a new answer and is distinct from reusing an EDSL response.

Inputs were validated with `ep validate -file ...` before execution. The initial setup attempt is excluded from the analysis. The conjunction family was saved before an export-code error; its checkpoint was reused when the remaining families resumed under the same inference configuration. This was a local exploratory workflow, not a preregistered study.

4 Experimental design and scoring

The unit of observation is one model completion for one question. The battery contains 11 families, 21 conditions, and four completions per condition. No prior interview response was included in subsequent prompts. Inspection of saved prompts confirmed an empty system prompt and an empty default agent trait dictionary. Bias labels and scoring keys were retained as metadata but were not rendered into the question. All questions appended a request for a brief explanation; EDSL then appended its answer-format instructions.

For multiple choice, forward and reversed option orders each received two draws. This controls a simple first-versus-last preference imperfectly: for three- and four-option questions, it is not full positional counterbalancing. In particular, the three-option decoy’s middle alternative remains in the middle on reversal. No free-text versus forced-choice comparison was run.

The analysis separates answer-key tasks from preference comparisons. Conjunction, Bayes-rule, independent-coin, monetary sunk-cost, logical-card, and arithmetic items have explicit keys. Numerical responses were scored within 0.1 percentage points for Bayes-rule items and 10^{-6} for arithmetic items. The other families do not define a universally correct preference. For binary choices, their descriptive contrast is

$$\hat{\Delta} = \frac{1}{4} \sum_{r=1}^4 I(Y_{r,1} = y^*) - \frac{1}{4} \sum_{r=1}^4 I(Y_{r,0} = y^*),$$

where y^* is the relevant option and the conditions differ in framing, default, timing, or choice set. These are contrasts between independent prompts, not within-person preference reversals. With four draws per cell, reported frequencies change in 25-percentage-point increments. No null-effect or statistical-significance claim is made.

The base-rate answer key follows directly from Bayes’ rule. With prevalence p , sensitivity 0.8, and false-positive rate 0.1,

$$P(D \mid F) = \frac{0.8p}{0.8p + 0.1(1 - p)}.$$

This gives 7.48% at $p = 0.01$ and 84.21% at $p = 0.40$. The project-stopping key follows from an incremental payoff of $2500 - 4000 = -\$1500$; the prompt explicitly excludes reputation, learning,

and other consequences. The arithmetic keys solve $x + (x + 8) = 8.8$ and $w = 3 + w/2$.

5 Results

5.1 Answer-key tasks

The model matched the specified key in 41 of 44 scored responses (93.2%). All three errors arose in the new conjunction vignette. This fraction is a summary of a selected battery, not an estimate of general rationality or a population-level bias rate. Table 2 reports the underlying counts.

Table 2: Tasks with explicit answer keys.

Task	Correct answer or rule	Correct	Total
Linda conjunction	Bank teller alone	4	4
New conjunction	Accountant alone	1	4
Base rates	7.48% and 84.21%	8	8
Independent coin	Heads and tails equally likely	8	8
Sunk cost	Stop the project	8	8
Logical falsification	Turn over E and 7	4	4
Intuitive arithmetic	Pen: \$0.40; cake: 6 kg	8	8
Total		41	44

The conjunction contrast merits attention. For Linda, the model always chose “Linda is a bank teller.” For Morgan, described as interested in astronomy, it selected “Morgan works as an accountant and belongs to an astronomy club” in three of four trials. An event $A \cap B$ is a subset of A , so $P(A \cap B) \leq P(A)$ regardless of how strongly the biography supports B . One final explanation stated:

“His astronomy interests make the additional club membership more likely.”

That reasoning concerns the plausibility of the added attribute but does not establish the required probability comparison. The error appeared under both option orders (Figure 1). The difference is consistent with uneven transfer across surface forms; it neither proves training-set memorization nor isolates familiarity from other wording differences.

The remaining keyed tasks were uniformly correct. The model used the prevalence in its scanner calculations, respected explicitly stated coin-flip independence, ignored past expenditure in the stipulated net-money problem, identified the cards capable of falsifying the conditional rule, and avoided both arithmetic lures. These tasks frequently supplied assumptions that make the normative solution especially explicit. Success on the card item should not be generalized to motivated reasoning or confirmation bias in richer information-search settings.

5.2 Preference and estimation contrasts

Table 3 reports all remaining families. None showed a clear bias-direction shift in these samples. That finding is narrower than evidence of bias immunity. The decoy test, for example, had a 100% bundle choice rate before the decoy was added, leaving no upward room for the classic attraction

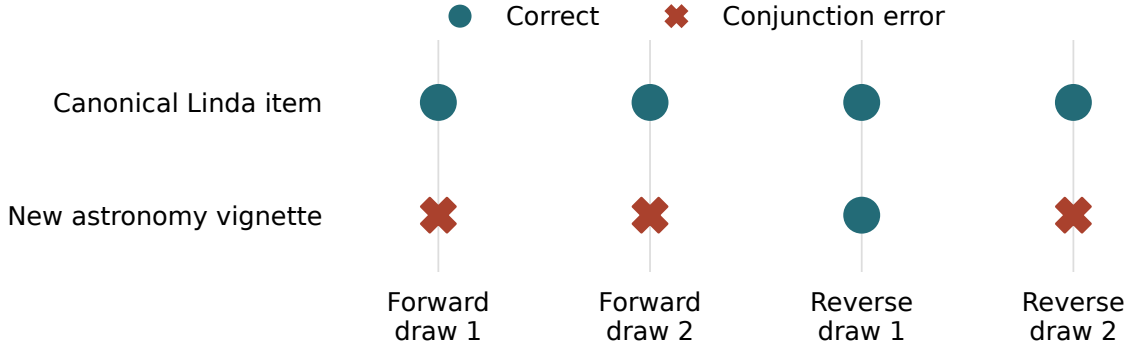


Figure 1: Every conjunction response, grouped by option order and iteration. The horizontal positions identify sampling conditions, not a chronological sequence. Circles are correct answers; crosses are conjunction errors. Four draws per vignette support a descriptive contrast, not a precise error-rate estimate.

effect. The anchoring prompt explicitly labeled the number as random and supplied three informative comparison projects.

Table 3: Matched-condition results; four responses per condition.

Family	Observed choices or estimates	Interpretation
Framing	Certain recovery: 4/4 in saved frame; 4/4 in lost frame	No observed shift; certainty preference is not itself an error.
Anchoring	Mean hours: 42.92 after anchor 10; 42.77 after anchor 100	Difference -0.15 hours; rounding-sized and opposite to assimilation.
Defaults	Plan B: 2/4 when A preselected; 1/4 when B preselected	Difference -25 percentage points, opposite to default-following; highly imprecise.
Payment timing	Later \$60: 4/4 against \$50 today; 4/4 when both dates shifted a year	No observed present-bias pattern; not a within-person reversal test.
Decoy	Bundle: 4/4 without print-only option; 4/4 with it	No observed change; baseline ceiling prevents an upward shift.

For framing, the guaranteed recovery of 300 of 900 files was preferred to a one-third chance of recovering all files and a two-thirds chance of recovering none. Both descriptions imply the same outcomes and the same expected number of recovered files. For payment timing, the model always selected \$60 rather than \$50 separated by 30 days. This choice is consistent across the tested time shifts; it does not establish a particular discount function.

5.3 Response integrity and explanation quality

Every retained response passed EDSL’s answer validation and carried the normal `stop` finish reason. No completion was marked as stopped by the output token limit. However, a configured thinking budget can still curtail reasoning before a final answer; the finish marker does not imply unrestricted reasoning.

One default-condition completion placed an extended reasoning passage into the answer channel. EDSL parsed Plan A, and the final text after the closing thinking tag also said Plan A. That observation was retained and flagged, rather than counted as a cleanly separated final explanation.

Excluding it would leave Plan B at 2/3 with A preselected versus 1/4 with B preselected, still opposite to default-following; neither comparison is precise. A separate payment-timing explanation described the specified 30-day gap as “10-day.” Thus successful schema validation establishes an acceptable answer value, not factual accuracy, faithfulness, or consistency of its accompanying explanation. No explanation is treated as direct evidence of the model’s causal reasoning process.

6 Interpretation and next experiments

The evidence supports a modest characterization: the model often produced normatively correct answers under explicit assumptions, while its performance on conjunction reasoning was sensitive to the vignette. Calling it “fairly rational” is reasonable as an informal description of this selected sample, but not as a demonstrated general property.

Several design limitations matter. There is only one model, one runtime, one reasoning-budget configuration, and a small number of draws. The draws are not human respondents or independently trained models. Temperature-driven variation does not provide demographic variation. Some tasks are widely recognizable, while the “new” vignette is newly written for this exercise, not verified absent from training data. Several prompts explicitly state the assumptions or identify an irrelevant cue, which may reduce their difficulty. The decoy condition has a ceiling effect, response-position control is partial, and explanations introduce an additional elicitation choice. No human baseline, alternative model, or full-precision comparator was run.

A stronger follow-up would cross many novel vignettes with surface wording and option order, fix the scoring and analysis plan before inference, and increase draws per cell. The conjunction contrast should be replicated across domains and reasoning budgets before attributing it to familiarity. Preference tasks should first be calibrated to avoid floor and ceiling choice shares, then tested with less explicit cues. A model comparison would use the same prompts and record actual token use, latency, failure rates, and resource consumption. The present results do not identify a cost–quality frontier for deployment.

7 Macaw and Modal: a deployment assessment

The local implementation does not transfer unchanged to the current Macaw sandbox. Inspection of the `coopr` checkout on September 19 found `macaw/sandbox_v2/runtime.py` requesting one CPU and 2,048 MiB of RAM, with no GPU argument. Its per-session Modal volume is mounted at `/home/user`; the image is Linux-based. These are source defaults, not verified live deployment settings. The RAM request should not be interpreted as a measured hard ceiling. Nevertheless, a small CPU-oriented sandbox is not a demonstrated environment for efficient 27B-class inference, and its persistent volume supplies storage rather than inference compute.

A plausible architecture keeps Macaw’s session sandbox and adds a separate GPU-backed inference worker. Store weights once on a model volume, load them into the worker, and expose an authenticated OpenAI-compatible endpoint to EDSL. Modal documents both GPU allocation and persistent model-weight storage (Modal, 2026a,b). The macOS executable would be replaced with a compatible PrismML Linux/CUDA build; the API integration can retain the same service abstraction while changing the endpoint and authentication configuration.

An L4 is a reasonable *candidate for a pilot*, not a measured optimum. PrismML’s model card reports L4 generation throughput of 29.8 tokens/s for PQ2_0 and 32.1 tokens/s for PTQ1_0 under

its own batch-one benchmark (PrismML, 2026a). These are vendor measurements on specified kernels and packing formats, not results from this Macaw deployment. A pilot should verify CUDA compatibility, actual memory use at the desired context and concurrency, cold-start loading time, throughput, and billed cost. No Modal inference was deployed or benchmarked in this exercise, and the behavioral battery does not evaluate Bonsai as a replacement for Macaw’s full agent or tool-use model.

8 Conclusion

EDSL successfully orchestrated local Bonsai inference, validated structured answers, and preserved a reproducible record of a small behavioral battery. The system was practical on the tested MacBook Pro. Its empirical profile was strong on explicit numerical and logical questions but uneven across the two conjunction vignettes. The appropriate outcome is a tested local workflow and a concrete hypothesis for further evaluation, rather than a broad claim of rationality or deployment equivalence.

Preparation and data availability. This technical note was prepared with AI assistance from the local scripts and saved results. It is an exploratory report, not a peer-reviewed paper. Inputs, scoring rules, parsed responses, and full raw results remain in `examples/prismml_local/behavioral_bias_run/`. No new inference was performed to produce the report. The source bundle includes the LaTeX, figure, references, and exported condition/response data; the larger Git-backed EDSL packages stay in the experiment directory.

References

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A Reproduction and artifact map

The original model repository revision was

```
6ed5e12bf84b7a63069882c91dd9e9218647d17b.
```

The SHA-256 digest of the tested PQ2_0 weight file was

```
3907dc1658db1f78a9826bf8d5bcb8dc65db0d466388937af57f2294fae62ec1.
```

The runtime release was `prism-b10683-d8f26ee`. Download metadata, including the runtime archive digest, is stored in `model-downloads.json`. The protocol records inference parameters and hashes of the experimental scripts. Model-sampling randomness was not seeded, so identical reruns should not be expected to reproduce every answer exactly.

From the EDSL repository root, the following commands build the validated inputs, execute or resume the battery, and generate its data summaries:

```
python examples/prismml_local/behavioral_biases.py --build-only
python examples/prismml_local/behavioral_biases.py
python examples/prismml_local/summarize_biases.py
```

The runner reuses complete saved family results. A new experiment requires a new `-output` directory. The standalone summary script defaults to the original run directory; its `summarize(output)` function can be called with another path. Individual family Jobs require the correct iteration count when executed manually: two for multiple-choice families and four for numerical families. To inspect an existing Results package without inference:

```
ep results select \
  --file examples/prismml_local/behavioral_bias_run/conjunction.results.ep \
  --column scenario.condition --column answer.response
```

Table 4: Core reproducibility artifacts. Paths are relative to the experiment directory unless indicated.

File or pattern	Contents
<code>conditions.json</code>	All 21 prompts, alternatives, and answer keys
<code>protocol.json</code>	Parameters, ordering design, and source hashes
<code>*.jobs.ep</code>	Saved EDSL execution objects
<code>*.results.ep</code>	Raw responses, rendered prompts, answers, and validation
<code>responses.csv/json</code>	All 84 extracted response records
<code>summary.json</code>	Per-condition counts and completion audit
<code>../behavioral_biases.py</code>	Input construction, execution, checkpointing, and export
<code>../ask_france.py</code>	Managed server helper and editable single-question example
<code>../start-server.sh</code>	Local runtime invocation

B Complete prompt inventory and observed distributions

Each question below appended: “Give a brief explanation in at most two sentences.” Multiple-choice questions then used EDSL’s instructions to select one option string and optionally put a comment

on the next line. No earlier question was included. Options are listed in the forward order; reversed order was also used as described in the main text. Each displayed example explanation is the first available final explanation in the exported ordering, excluding the response with reasoning in the answer channel. Full draw-level records remain in the accompanying data files. These quotations are model outputs and may themselves contain errors or overly strong claims.

B.1 Conjunction: linda

Question. Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations. Which is more probable?

Options (forward order).

1. Linda is a bank teller.
2. Linda is a bank teller and is active in the feminist movement.

Observed answers. Linda is a bank teller. (4/4).

Scoring key. Linda is a bank teller..

Example final explanation. “The single statement is more probable because adding "active in the feminist movement" makes a narrower, less likely conjunction.”

B.2 Conjunction: new vignette

Question. Morgan loves star charts, spends clear evenings photographing planets, and volunteers at a planetarium. Which is more probable?

Options (forward order).

1. Morgan works as an accountant.
2. Morgan works as an accountant and belongs to an astronomy club.

Observed answers. Morgan works as an accountant. (1/4); Morgan works as an accountant and belongs to an astronomy club. (3/4).

Scoring key. Morgan works as an accountant..

Example final explanation. “The second option is more probable because it is a less specific claim than asserting both being an accountant and belonging to an astronomy club.”

B.3 Base rates: rare

Question. A factory produces parts, of which 1% are defective. A scanner flags 80% of defective parts and also flags 10% of non-defective parts. A part is randomly selected and is flagged. What is the probability that it is defective? Give a percentage from 0 to 100, rounded to two decimal places.

Observed answers. 7.48 (4/4).

Scoring key. 7.48.

Example final explanation. “Using Bayes’ rule: $(0.01 \times 0.80) / [(0.01 \times 0.80) + (0.99 \times 0.10)] \approx 0.074766$, or 7.48%.”

B.4 Base rates: common

Question. A factory produces parts, of which 40% are defective. A scanner flags 80% of defective parts and also flags 10% of non-defective parts. A part is randomly selected and is flagged. What is the probability that it is defective? Give a percentage from 0 to 100, rounded to two decimal places.

Observed answers. 84.21 (4/4).

Scoring key. 84.21.

Example final explanation. “ $P(\text{defective} \mid \text{flagged}) = (0.8 \times 0.4) / (0.8 \times 0.4 + 0.1 \times 0.6) = 0.8421$, which is 84.21%.”

B.5 Independent coin flips: streak

Question. A fair coin has been flipped six times, producing heads, heads, heads, heads, heads. Each flip is independent. On the next flip, which outcome is more likely?

Options (forward order).

1. Heads
2. Tails
3. Heads and tails are equally likely

Observed answers. Heads and tails are equally likely (4/4).

Scoring key. Heads and tails are equally likely.

Example final explanation. “Each independent fair coin flip has a 50/50 chance of heads or tails.”

B.6 Independent coin flips: mixed

Question. A fair coin has been flipped six times, producing heads, tails, tails, heads, tails, heads. Each flip is independent. On the next flip, which outcome is more likely?

Options (forward order).

1. Heads
2. Tails
3. Heads and tails are equally likely

Observed answers. Heads and tails are equally likely (4/4).

Scoring key. Heads and tails are equally likely.

Example final explanation. “Each flip of a fair coin is independent, so previous outcomes do not affect the next flip.”

B.7 Framing: saved

Question. A damaged archive contains 900 files. Choose a recovery plan. Plan A will save exactly 300 files. Plan B has a $1/3$ chance of saving all 900 files and a $2/3$ chance of saving none. Every file is equally valuable; these are the complete outcomes. Which plan do you choose?

Options (forward order).

1. Plan A
2. Plan B

Observed answers. Plan A (4/4).

Example final explanation. “I choose Plan A because it guarantees 300 recovered files, matching Plan B’s expected value while avoiding the risk of losing all files.”

B.8 Framing: lost

Question. A damaged archive contains 900 files. Choose a recovery plan. Plan A will lose exactly 600 files. Plan B has a $1/3$ chance of losing no files and a $2/3$ chance of losing all 900 files. Every file is equally valuable; these are the complete outcomes. Which plan do you choose?

Options (forward order).

1. Plan A
2. Plan B

Observed answers. Plan A (4/4).

Example final explanation. “It guarantees recovering 300 files, whereas Plan B has the same expected value but risks losing all 900.”

B.9 Anchoring: low

Question. You are estimating the engineering hours for a new module. Three comparable modules required 38, 44, and 46 hours. The new module is similar in scope and staffing. Before you estimate, an unrelated random-number generator displays 10. Consider whether the new module will take more or less than 10 hours. What is your best estimate of its total engineering hours? Return a number of hours.

Observed answers. 43 (3/4); 42.67 (1/4).

Example final explanation. “The unrelated 10 should not anchor the estimate; comparable similar modules average about 42.67 hours.”

B.10 Anchoring: high

Question. You are estimating the engineering hours for a new module. Three comparable modules required 38, 44, and 46 hours. The new module is similar in scope and staffing. Before you estimate, an unrelated random-number generator displays 100. Consider whether the new module will take more or less than 100 hours. What is your best estimate of its total engineering hours? Return a number of hours.

Observed answers. 42.67 (1/4); 42.7 (2/4); 43 (1/4).

Example final explanation. “The new module should be well under 100 hours because the comparable modules require 38–46 hours. I estimate the average of those values, 42.67 hours.”

B.11 Sunk cost: none

Question. You manage a project and have already spent \$0, which cannot be recovered. Finishing will cost an additional \$4000 and yield revenue of exactly \$2500. Stopping costs nothing and yields no further revenue. There are no reputation, learning, contractual, or other consequences. Your sole objective is to maximize final net money. What do you do?

Options (forward order).

1. Finish the project
2. Stop the project

Observed answers. Stop the project (4/4).

Scoring key. Stop the project.

Example final explanation. “Stopping avoids an additional net loss of \$1500.”

B.12 Sunk cost: large

Question. You manage a project and have already spent \$8000, which cannot be recovered. Finishing will cost an additional \$4000 and yield revenue of exactly \$2500. Stopping costs nothing and yields no further revenue. There are no reputation, learning, contractual, or other consequences. Your sole objective is to maximize final net money. What do you do?

Options (forward order).

1. Finish the project
2. Stop the project

Observed answers. Stop the project (4/4).

Scoring key. Stop the project.

Example final explanation. “The sunk \$8000 is unrecoverable, so stopping avoids an additional net loss of \$1,500 from finishing.”

B.13 Logical falsification: card test

Question. Each card has a letter on one side and a number on the other. Four cards show E, K, 4, and 7. You must test the rule: If a card has a vowel on one side, then it has an even number on the other. Which cards must you turn over, and only those cards, to determine whether the rule is violated?

Options (forward order).

1. E and 4
2. E and 7
3. E only
4. All four cards

Observed answers. E and 7 (4/4).

Scoring key. E and 7.

Example final explanation. “Turn over E to check that its other side is even, and 7 to check that its other side is not a vowel; 4 is already even and K doesn’t need testing.”

B.14 Defaults: a default

Question. Choose between two internet plans. Plan A costs \$40 per month with 100 Mbps service. Plan B costs \$50 per month with 200 Mbps service. All other terms are identical. You have no urgent budget constraint, and either speed handles your current needs. The form randomly preselects Plan A; this is not a recommendation. Choosing either plan takes the same effort and has no switching cost. Which do you choose?

Options (forward order).

1. Plan A
2. Plan B

Observed answers. Plan A (2/4); Plan B (2/4).

Example final explanation. “Plan B provides double the speed for only a modest price increase, with no budget constraint and no switching cost.”

B.15 Defaults: b default

Question. Choose between two internet plans. Plan A costs \$40 per month with 100 Mbps service. Plan B costs \$50 per month with 200 Mbps service. All other terms are identical. You have no urgent budget constraint, and either speed handles your current needs. The form randomly preselects Plan B; this is not a recommendation. Choosing either plan takes the same effort and has no switching cost. Which do you choose?

Options (forward order).

1. Plan A
2. Plan B

Observed answers. Plan A (3/4); Plan B (1/4).

Example final explanation. “It costs less and still meets your needs.”

B.16 Payment timing: now

Question. Choose one guaranteed payment: \$50 today, or \$60 in 30 days. Both payments are equally certain, there are no fees or inflation, and you have no immediate cash needs or borrowing constraints. Which do you choose?

Options (forward order).

1. The earlier \$50 payment
2. The later \$60 payment

Observed answers. The later \$60 payment (4/4).

Example final explanation. “The guaranteed \$60 is equivalent to a 20% return over 30 days, which I would accept given no immediate cash needs.”

B.17 Payment timing: later

Question. Choose one guaranteed payment: \$50 in 365 days, or \$60 in 395 days. Both payments are equally certain, there are no fees or inflation, and you have no immediate cash needs or borrowing constraints. Which do you choose?

Options (forward order).

1. The earlier \$50 payment
2. The later \$60 payment

Observed answers. The later \$60 payment (4/4).

Example final explanation. “The later \$60 payment is worth more in nominal terms, and with certainty, no borrowing, no costs, and no cash need, the extra \$10 outweighs the 30-day delay.”

B.18 Decoy: without decoy

Question. Choose an annual magazine subscription for yourself. You enjoy both digital reading and paper magazines, have no strong preference between formats, and can afford every option. All options contain the same editorial content. Which subscription do you choose?

Options (forward order).

1. Digital subscription: \$60
2. Digital and print subscription: \$120

Observed answers. Digital and print subscription: \$120 (4/4).

Example final explanation. “I choose it because it supports both my digital and paper reading preferences without any preference between formats and I can afford it.”

B.19 Decoy: with decoy

Question. Choose an annual magazine subscription for yourself. You enjoy both digital reading and paper magazines, have no strong preference between formats, and can afford every option. All options contain the same editorial content. Which subscription do you choose?

Options (forward order).

1. Digital subscription: \$60
2. Digital and print subscription: \$120
3. Print-only subscription: \$120

Observed answers. Digital and print subscription: \$120 (4/4).

Example final explanation. “I choose this because it includes both formats I enjoy at the same price as print-only.”

B.20 Intuitive arithmetic: notebook pen

Question. A notebook and a pen cost \$8.80 in total. The notebook costs \$8.00 more than the pen. How much does the pen cost, in dollars?

Observed answers. 0.4 (4/4).

Scoring key. 0.4.

Example final explanation. “Let p be the pen cost; then $p + (p + 8) = 8.80$, so $2p = 0.80$ and $p = 0.4$.”

B.21 Intuitive arithmetic: cake

Question. A whole cake weighs 3 kilograms plus half of its own weight. How many kilograms does the whole cake weigh?

Observed answers. 6 (4/4).

Scoring key. 6.

Example final explanation. “Let W be the cake’s weight; $W = 3 + W/2$, so $W = 6$.”